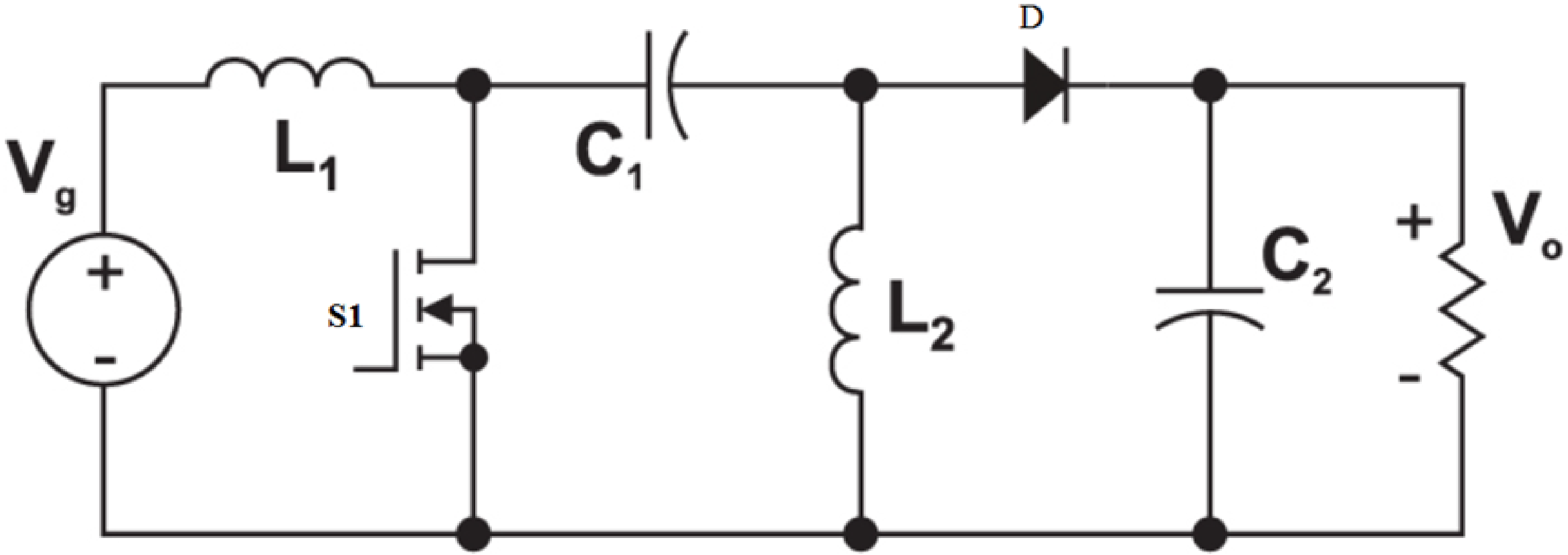
SEPIC converter

Dc-Dc converter

SEPIC or single-ended primary inductor converter is a type of Buck/Boost DC-DC Converter that provides a DC output greater than, less than, or equal to its DC input. It consists of two inductors, one is at the input and another one is connected to the ground and these two inductors are connected by a coupling capacitor.



The major advantage of a SEPIC converter is that its output voltage has the same polarity as that of the input voltage. This is called no-polarity inversion.

SEPIC converter circuit consists of input voltage source Vg, coupling capacitor C1, and output capacitor C2, two inductors L1 and L2, diode D, and load resistance. SEPIC converter circuits share the common ground between the input and output circuits. This converter exchanges energy between the inductor L1, capacitor C1, and inductor L2 to convert input DC voltage to require output voltage level. Typically, a power transistor switch (S1) such as a MOSFET is used to control the amount of energy exchanged.

Instead of using discrete inductors (L1 & L2), the SEPIC converter can also be designed using the coupling inductors which will improve the efficiency and reduce PCB area. Coupling inductors mean that two windings (L1 & L2) are connected in a single core.

Advantages of SEPIC Converter :

- In buck-boost and Cuk converters the polarity of the output voltage will be opposite of the input voltage i.e., an inverted output voltage will be obtained. Whereas in SEPIC the polarity of the output voltage will be the same as the input voltage.
- The capacitor C1 will isolate the input and output sides from each other i.e., an abnormal condition occurring on any side of the circuit will not affect the other side.
- Reduced input current ripple.
- Cuk and buck-boost converter operation cause a large amount of electrical stress on the components which can be overcome in SEPIC.
- High efficiency and stable operation.
- The use of coupled inductor instead of two inductors (two inductors wound onto the single core) makes the circuit compact.

Disadvantages of SEPIC Converter :

- Similar to the buck-boost converter, the output current of SEPIC can be pulsating. Whereas in the Cuk converter the output current is continuous.

- All the current from the source to load has to go through capacitor C1, thus a capacitor of high capacitance and current handling capability is required.

- Since the voltage across the capacitor C1 reverses for every cycle it should be of a non-polarized type.

Applications of SEPIC Converter :

- DC power supply systems with unregulated inputs.
- Battery-operated equipment.
- Lighting applications.

Given parameters:

$V_s = 9\ V$

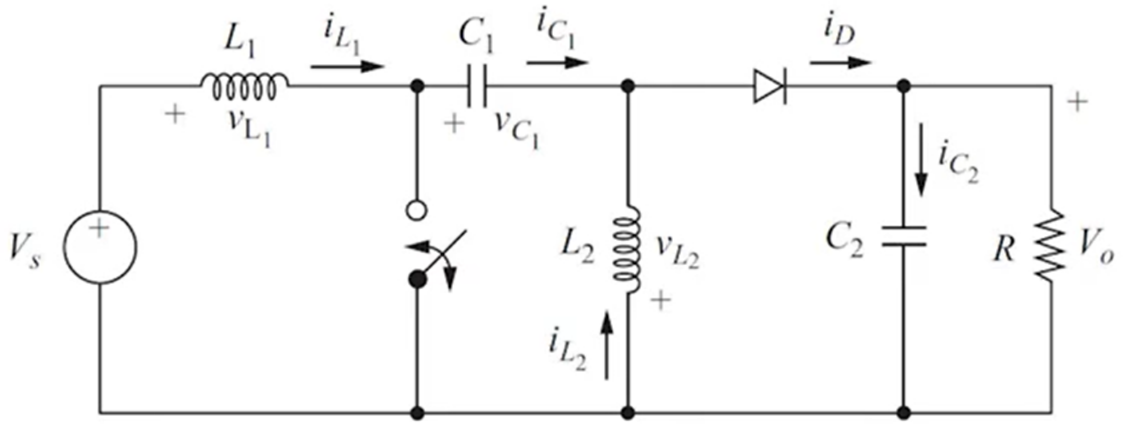
$V_o = 6\ V$

$\Delta i_{L_1} = \Delta i_{L_2} = 0.4$

$\Delta v_{C_1} = \Delta v_{C_2} = 0.1$

$f = 100000\ Hz$

$Load = 12\ \mathrm{W}$



SEPIC
dc to dc
converter

1. Calculate duty ratio

$$D = \frac{V_o}{V_o + V_s} = \frac{6}{6+9} = 0.4$$

2. Calculate average inductance current I_{L_1}

$$I_{L_1} = \frac{P_s}{V_s} = \frac{12}{9} = 1.33\ A$$

3. Calculate average inductance current I_{L_2}

$$I_{L_2} = \frac{P_s}{V_o} = \frac{12}{6} = 2\ A$$

4. Calculate maximum inductance current $I_{L_{1,max}}$

$$I_{L_{1,max}} = I_{L_1} + \frac{\Delta i_{L_1}}{2} = 1.33 + \frac{0.4}{2} = 1.53\ A$$

5. Calculate minimum inductance current $I_{L_{1,min}}$

$$I_{L_{1,min}} = I_{L_1} - \frac{\Delta i_{L_1}}{2} = 1.33 - \frac{0.4}{2} = 1.13\ A$$

6. Calculate maximum inductance current $I_{L_{2,max}}$

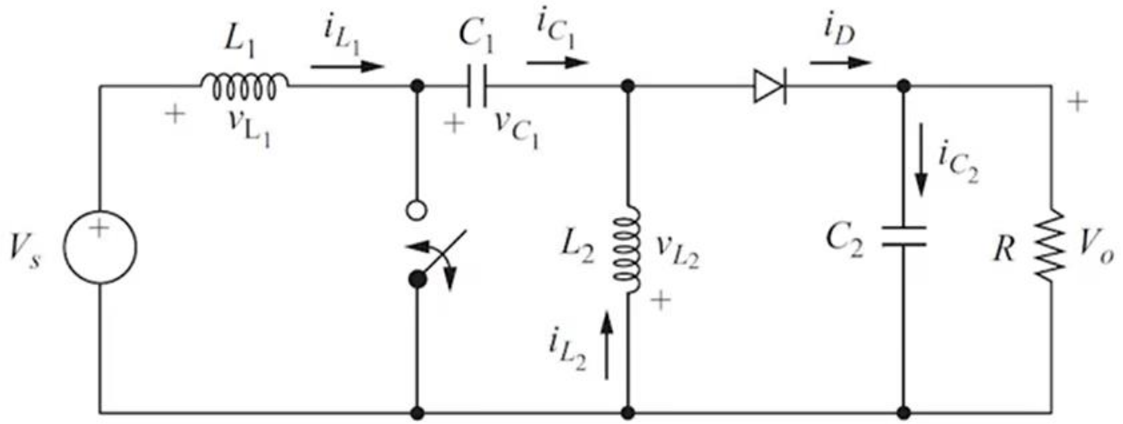
$$I_{L_{2,max}} = I_{L_2} + \frac{\Delta i_{L_2}}{2} = 2 + \frac{0.4}{2} = 2.2A$$

Given parameters:

$V_s = 9\,V$ $\qquad \Delta v_{C_1} = \Delta v_{C_2} = 0.1$

$V_o = 6\,V$ $\qquad f = 100000\,Hz$

$\Delta i_{L_1} = \Delta i_{L_2} = 0.4$ $\qquad Load = 12\text{ W}$



SEPIC dc to dc converter

7. Calculate minimum inductance current $I_{L_{2,min}}$

$$I_{L_{2,min}} = I_{L_2} - \frac{\Delta i_{L_2}}{2} = 2 - \frac{0.4}{2} = 1.8\,A$$

8. Calculate inductance L_1

$$L_1 = \frac{V_s D}{f \Delta i_{L_1}} = \frac{(9)(0.4)}{(100000)(0.4)} = 90\,\mu\text{H}$$

9. Calculate inductance L_2

$$L_2 = \frac{V_s D}{f \Delta i_{L_2}} = \frac{(9)(0.4)}{(100000)(0.4)} = 90\,\mu\text{H}$$

10. Calculate load resistance

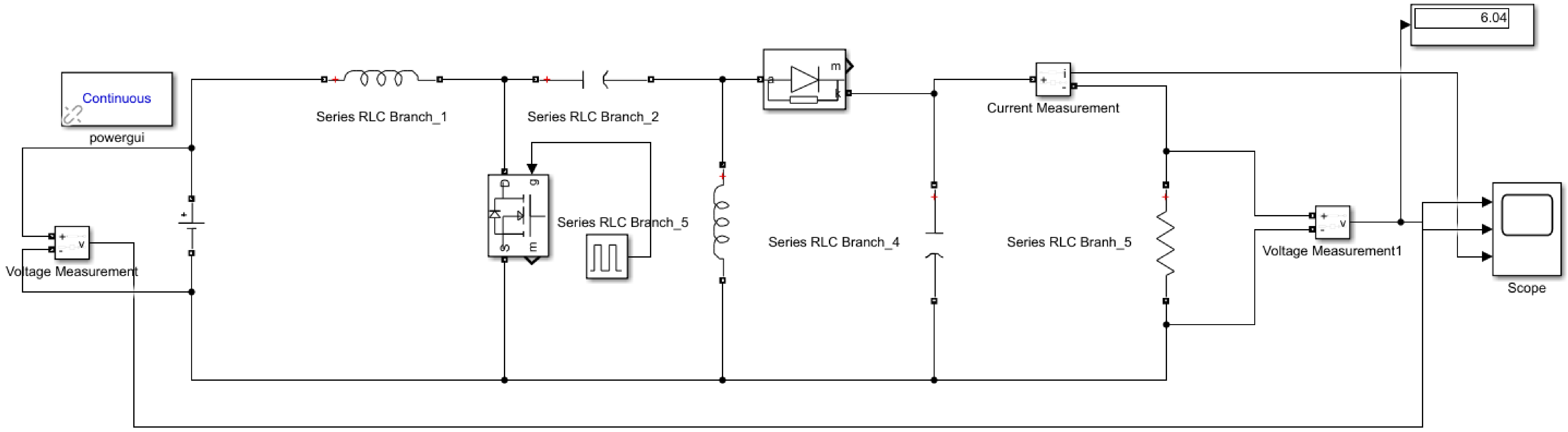
$$R = \frac{V_o^2}{P} = \frac{(6)^2}{12} = 3\,\Omega$$

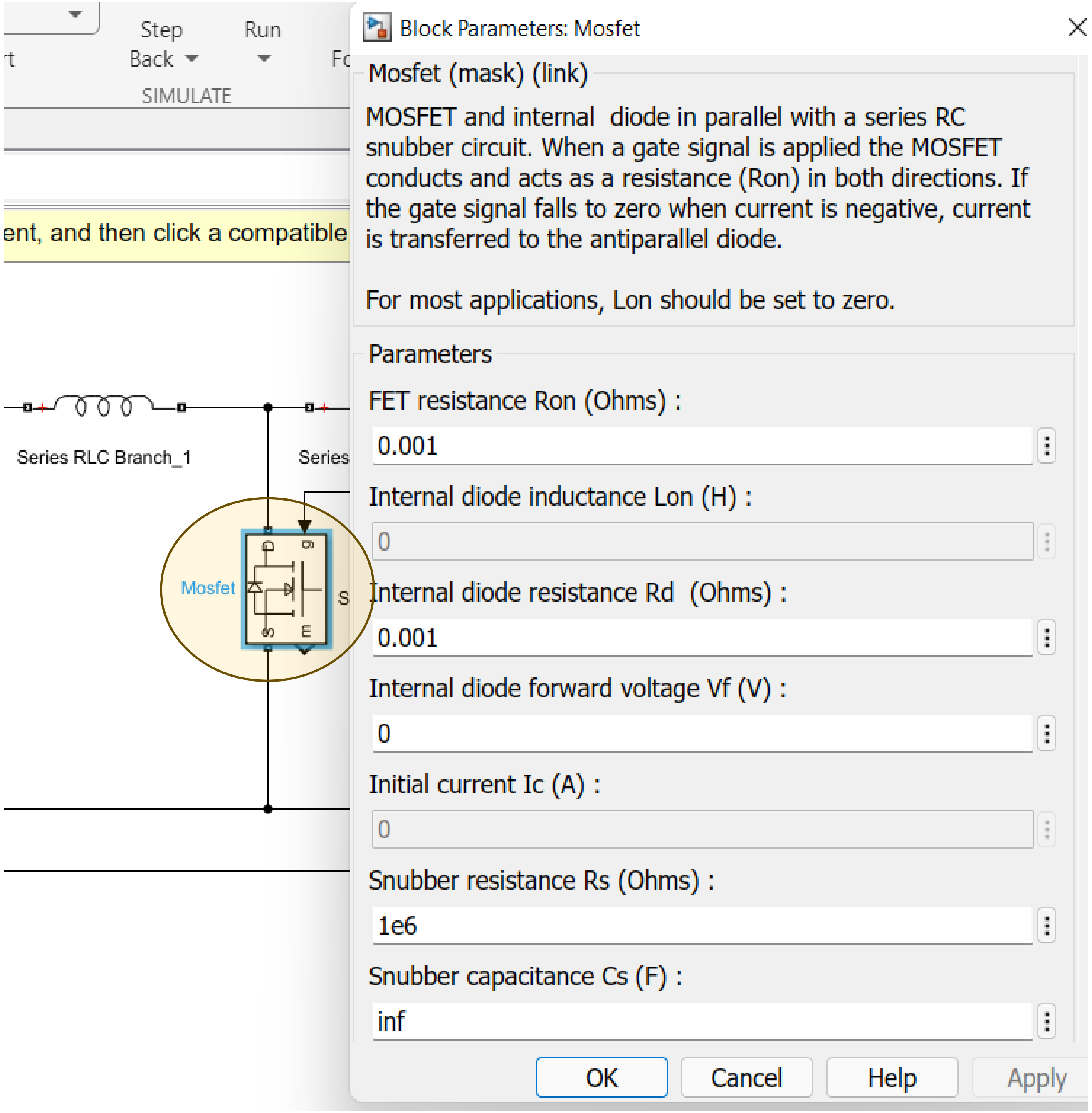
11. Calculate capacitance C_1

$$C_1 = \frac{V_o D}{R f \Delta v_{C_1}} = \frac{(6)(0.4)}{(3)(100000)(0.1)} = 80\,\mu\text{F}$$

12. Calculate capacitance C_2

$$C_2 = \frac{V_o D}{R f \Delta v_{C_2}} = \frac{(6)(0.4)}{(3)(100000)(0.1)} = 80\,\mu\text{F}$$


Continuous
powergui
Series RLC Branch_1
Series RLC Branch_2
m
a
k
6.04
Current Measurement
G
g
S
E
Series RLC Branch_5
Series RLC Branch_4
Series RLC Branh_5
Voltage Measurement
Voltage Measurement1
Scope


Step
Back
Run
SIMULATE
ent, and then click a compatible
Series RLC Branch_1
Series
Mosfet
Block Parameters: Mosfet
Mosfet (mask) (link)
MOSFET and internal diode in parallel with a series RC snubber circuit. When a gate signal is applied the MOSFET conducts and acts as a resistance (Ron) in both directions. If the gate signal falls to zero when current is negative, current is transferred to the antiparallel diode.
For most applications, Lon should be set to zero.
Parameters
FET resistance Ron (Ohms) :
0.001
Internal diode inductance Lon (H) :
0
Internal diode resistance Rd (Ohms) :
0.001
Internal diode forward voltage Vf (V) :
0
Initial current Ic (A) :
0
Snubber resistance Rs (Ohms) :
1e6
Snubber capacitance Cs (F) :
inf
OK
Cancel
Help
Apply